YIBO LIN

Associate Professor ◇ Department of Design Automation and Computing System
School of Integrated Circuits, 5 Yiheyuan Road, Beijing, China, 100871 ◇ Peking University
yibolin@pku.edu.cn ◇ www.yibolin.com

RESEARCH INTERESTS

Modeling and optimization in VLSI CAD, machine learning applications, and heterogeneous computing

EDUCATION

University of Texas at Austin, Texas, USA Aug. 2013 – May 2018
Ph.D., Department of Electrical and Computer Engineering

Shanghai Jiao Tong University, Shanghai, China Sep. 2009 – Jun. 2013
B.S., Department of Microelectronics

EXPERIENCE

Peking University, Beijing, China Aug. 2025 – present
Associate Professor

Peking University, Beijing, China Jul. 2019 – Jul. 2025
Assistant Professor
Department of Design Automation and Computing System,
School of Integrated Circuits (since Nov. 2021)
Center for Energy-efficient Computing and Applications (CECA),
School of Electronics Engineering and Computer Science

University of Texas at Austin, Texas, U.S. Jun. 2018 – Jun. 2019
Postdoc
Department of Electrical and Computer Engineering

TEACHING EXPERIENCE

Practice in AI-Assisted VLSI Circuit Design		
Instructor	Graduate	2026
Fundamental Algorithms for Engineering in Integrated Circuits		
Instructor	Graduate	2023–2026
Design Automation and Computing System		
Instructor	Graduate	2022–2024
Optimization and Machine Learning in VLSI Design Automation		
Instructor	Undergraduate	2021–2024
Introduction to Computing B		
Instructor	Undergraduate	2020–2022

AWARDS AND HONORS

DAC Under-40 Innovators Award	DAC	2026
Best Paper Award Nomination (13/544)	DAC	2026
Best Paper Award Nomination (×3) (12/145)	ISEDA	2026
Best Paper Award Nomination (19/266)	ICCAD	2025
Best Paper Award Nomination (7/420)	DAC	2025
Honorable Mention (×2) (/135)	ISEDA	2025
Best Paper Award Nomination (13/168)	ASPDAC	2025
Best Paper Award Nomination (10/195)	ICCAD	2024
Best Paper Award (6/151) & Honorable Mention	ISEDA	2024
Early Career Award (only one per year)	CCF Technical Committee in IC	2023
Inaugural Best Reviewer Award	ICCAD	2023
Best Paper Award (4/205)	DATE	2023
Best Paper Award (4/249)	DATE	2022
Best Paper Award Nomination (6/132)	ICCAD	2022
Donald O. Pederson Best Paper Award (2/3495 in 4 years)	TCAD	2021
Best Paper Award	ISPD	2020
Best Paper Award Nomination (12/86)	ASPDAC	2020
Best Paper Award (1/201) & Nomination (5/201)	DAC	2019
Best Paper Award Nomination	ISPD	2019
Inaugural Best Paper Award	Integration, the VLSI Journal	2018
Franco Cerrina Memorial Best Student Paper Award	SPIE	2016
A. Richard Newton Young Student Fellow	DAC	2014

PROFESSIONAL SERVICE

Executive Committee Member

- General co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2025–2026
- Program co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2024
- Panel chair, IEEE International Symposium of Electronics Design Automation (ISEDA) 2023–2024
- Financial chair, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD) 2021–2023

Journal Editor

- Associate Editor, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024 – present
- Associate Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES), 2024 – present
- Associate Editor, Elsevier Integration, the VLSI Journal (Integration), 2024 – present
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2024
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2022

Contest Chair

- Contest Problem Chair, Interated Circuits EDA Elite Challenge 2021 – 2023

Technical Program Committee Member

- ACM/IEEE Design Automation Conference (DAC): 2020
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2018 – 2021, 2023
- Design, Automation and Test in Europe Conference (DATE): 2025
- IEEE International Conference on Computer Design (ICCD): 2019
- IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC): 2021 – 2022
- ACM International Symposium on Physical Design (ISPD): 2020, 2025
- ACM/IEEE Workshop on Machine Learning for CAD (MLCAD): 2021
- ACM Great Lakes Symposium on VLSI (GLVLSI): 2024
- Workshop on Synthesis And System Integration of Mixed Information technologies (SASIMI): 2021
- IEEE Electron Devices Technology and Manufacturing Conference (EDTM): 2021
- IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS): 2022.

Journal Reviewer

- IEEE Transaction on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Computers (TC)
- ACM Transaction on Design Automation of Electronic Systems (TODAES)
- SPIE Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3)
- Elsevier, Integration, the VLSI Journal (Integration)

Volunteer

- ACM SIGDA Website Administrator, 2021 – present

10 SELECTED PUBLICATIONS

Inverse chronological order. * denotes corresponding authors.

1. Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin***, "Stronger Mixed-Size Placement Backbone Considering Second-Order Information", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.

2. Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)

3. Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin***, "Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2023.

4. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2023.

5. Qipan Wang, Xiaohan Gao, **Yibo Lin***, Runsheng Wang and Ru Huang, "DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)

6. Zizheng Guo and **Yibo Lin***, "Differentiable-Timing-Driven Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

7. Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)

8. Zizheng Guo, Jing Mai and **Yibo Lin***, "Ultrafast CPU/GPU Kernels for Density Accumulation in Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

9. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

10. **Yibo Lin***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, "DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

PUBLICATIONS

Book Chapters

[B3] **Yibo Lin**, Zizheng Guo and Jing Mai, "Deep Learning Framework for Placement", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

[B2] Haoyu Yang, **Yibo Lin** and Bei Yu, "Machine Learning for Mask Synthesis and Verification", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

Conference and Journal Papers (* denotes corresponding authors)

Summary: DAC (37), ICCAD (32), IEEE TCAD (34), DATE (21), etc.

[J251] Zhenkun Lin, **Yibo Lin**, Genggeng Liu* and Gang Du, "URoute: Universal Routability Prediction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J250] Xu He*, Yushan Wang, Chenjing Yang, Renjun Zhao, **Yibo Lin**, Peiyu Liao, Bei Yu, Yao Wang, Chang Liu and Yang Guo, "Efficient Timing Prediction and Optimization Using Derivable Gradient Boosting Machine Model at Placement Stage", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[J249] Chunyuan Zhao, Zizheng Guo, Zuodong Zhang and **Yibo Lin***, "GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J248] Xufeng Yao, Jiaxi Jiang, Yuxuan Zhao, Peiyu Liao, **Yibo Lin** and Bei Yu*, "EvoPlace: Evolution of Optimization Algorithms for Global Placement via Large Language Models", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J247] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "M2MOR: Multi-scale Thermo-mechanical Coupling Simulation of Chiplet Heterogeneous Integration Systems Based on Model Order Reduction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J246] Haoyi Zhang, Jiechen Huang, Bingyang Liu, Wenjian Yu*, **Yibo Lin*** and Runsheng Wang, "APEC-Route: Guiding Analog Routing with On-the-Fly Parasitic Extraction Based on a Neural Capacitance Model", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[C245] Mengjia Dai, Hongduo Liu, Yuhao Ji, Xiaojing Su, **Yibo Lin** and Bei Yu*, "Soft-Constrained Triple Patterning Layout Decomposition", IEEE International Conference on Computer Design (ICCD), Hong Kong, China, Nov 16-18, 2026.

[C244] Xiang Li, Zuodong Zhang and **Yibo Lin***, "TAT: Timing-Aware Steiner Tree Generation for Buffer Insertion", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C243] Yuxi Lin, **Yibo Lin**, Zepeng Li, Weijie Fang, Guangyong Chen and Genggeng Liu*, "Clock Tree Synthesis with Cross-Tier Skew and Wirelength Co-Optimization for Face-to-Face Hybrid Bonding 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C242] Zhenkun Lin, Taoyu Zhou, Qipan Wang, Tianxiang Zhu, **Yibo Lin**, Meng Li, Runsheng Wang, Yandong He and Gang Du*, "VULCAN: Unified Multi-Scale Chip Thermal Simulation System for Design-Technology Co-Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C241] Jingxin Wang, Sinian Zhang, Yun Liang, **Yibo Lin**, Pengpeng Ren, Runsheng Wang and Weikang Qian*, "GUALO: A Generalizable Uncertainty-Aware AI Agent for Logic Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C240] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Boyang Wu, Yuhao Ji, Chunyuan Zhao, Xinyu Qu, Yuxiang Zhao, Runzhe Tao, Bei Yu, Zhixiong Di and **Yibo Lin***, "LEGO 2.0: A Unified Agentic Platform for Digital Chip Design and Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C239] Weijian Fan, Zhiyuan Zhao, Jinpeng Wang, Haoyi Zhang, Jincheng Lou, Runsheng Wang, Xiyuan Tang and **Yibo Lin***, "VinePlace: Compact Analog and Mixed-Signal Placement via Core-Centered Vine Representation", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C238] Yuhao Ji, Siying Yu, Yunqi Shi, Fangzhou Liu, Yuxuan Zhao, Jiaxi Jiang, Zizheng Guo,

Chao Qian, **Yibo Lin** and Bei Yu*, "HeteroOpt3D: Timing Optimization for Heterogeneous Hybrid-bonding-based Face-to-Face 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C237] Kairong Guo, Haoyi Zhang, Haichuan Liu, Chunyuan Zhao and **Yibo Lin***, "ScoutCell: Sensitivity-Guided Coupling Optimization for Timing-Driven Standard Cell Synthesis", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026.

[C236] Zixuan Pan, Zizheng Guo, Yufan Du, Runsheng Wang and **Yibo Lin***, "Disentangled Differentiable Timing-Power Co-Optimization with Quad-Gradient Gate sizing", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Best Paper Nomination**)

[C235] Yi Ren, Yukun Wang, Xiang Meng, Guoyao Cheng, Baokang Peng, Lining Zhang, **Yibo Lin**, Runsheng Wang and Guangyu Sun*, "CellE: Automated Standard Cell Library Extension via Equality Saturation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C234] Jing Mai, Yi-Chen Lu, **Yibo Lin** and Haoxing Ren, "LATTE: Legality-Assured Differentiable Timing-Driven Detailed Placement", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C233] Wenbo An, Kairong Guo, Haoyi Zhang and **Yibo Lin***, "EPiCell: Electro-Physical Co-Modeling for Standard Cell PPA Prediction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C232] Yuhao Ji, Yunqi Shi, Tianshu Hou, Yuxuan Zhao, Chunyuan Zhao, Peiyu Liao, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu*, "Integrated Timing-driven Placement for Hybrid-Bonding-based Face-to-Face 3D ICs", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C231] Kexing Zhou, Youwei Zhuo, **Yibo Lin**, Weikang Qian, Pengpeng Ren and Yun Liang*, "Activity-Aware Partitioning for Effective Multi-Threaded Event-Driven RTL Simulation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C230] Jiechen Huang, Hector R. Rodriguez, Dingcheng Yang, Zuochang Ye, **Yibo Lin** and Wenjian Yu*, "AttentionCap: Transformer Based Capacitance Matrix Learning Toward Full-Chip Extraction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C229] Haoyi Zhang, Weijian Fan, Xiaohan Gao, Bingyang Liu, Runsheng Wang and **Yibo Lin***, "PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Invited Paper**)

[C228] Haoyi Zhang, Kairong Guo, Bojie Zhang, **Yibo Lin*** and Runsheng Wang, "FusionCell: Cross-Attentive Fusion of Layout Geometry and Netlist Topology for Standard-Cell Performance Prediction", International Conference on Machine Learning (ICML), Seoul, Korea, Jul 6-11, 2026.

[C227] Xizhe Shi, Zizheng Guo, Zuodong Zhang, Yun Liang, Runsheng Wang* and **Yibo Lin***, "HeteroPower: A CPU-GPU Heterogeneous Engine to Accelerate Gate-Level Power Analysis",

IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C226] Kairong Guo, Jiechen Huang, Wenjian Yu and **Yibo Lin***, "CAPCell: Standard Cell Layout Synthesis with Parasitic Capacitance Aware Parallel Sampling", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C225] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Runzhe Tao, Xinyu Qu and **Yibo Lin***, "LEGO: An LLM Skill-Based Front-End Design Generation Platform", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C224] Zizhuo Fu, Yifan Zhou, Zhaoxin Lu, Guangyu Sun, Runsheng Wang, Meng Li* and **Yibo Lin***, "RePart: Efficient Hypergraph Partitioning with Logic Replication Optimization for Multi-FPGA System", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C223] Xun Jiang, Haoran Lu, Yifan Chen, Jiarui Wang, Chunyuan Zhao, Yibo Zhang, Jianxiang Jin, Heng Wu, Runsheng Wang and **Yibo Lin***, "Flexible Double-Side Pin Redistribution with Efficient Virtual-Net-Guided Cell Type Assignment", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)

[C222] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "ElmoreCeff: A GPU-Friendly Elmore-Like Delay Calculator with a Closed-Form Effective Capacitance Model", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C221] Bingyang Liu, Haoning Jiang, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, David Z. Pan and **Yibo Lin***, "GRAIN: A Design-Intent-Driven Analog Layout Migration Framework", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C220] Yuxiang Zhao, Jing Mai, Zishu Li, Haoyi Zhang, Jincheng Lou, Qing He, Runsheng Wang and **Yibo Lin***, "PowerCube: A Formula-in-the-Loop Framework for Cross-Stage Power Map Prediction", IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.

[C219] Mingjun Wang, Yihan Wen, Yuntao Lu, Fengrui Liu, Yuxiang Zhao, Boyu Han, Jianan Mu, **Yibo Lin**, Runsheng Wang, Huawei Li and Bei Yu, "CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design", International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 23-27, 2026.

[C218] Chenglin Ye, Yuze Lu, Yizhan Liu, Ligong Zhang, Jinghan Xu, Fei Liu, **Yibo Lin**, Zheng Zhou and Xiaoyan Liu, "EMAPER: Cross-Level Electromigration Aware Placement and Routing EDA Workflow for Interconnects Hotspot Prediction and Mitigation", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C217] Jiajun Tan, Qichao Ma, Yiming Du, Yiming Gan, Ling Liang, **Yibo Lin**, Ming Zhu, Zongwei Wang and Yimao Cai, "SONIC: Smart Optimization for Neural-Integrated CMP with Timing-Aware Fills", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C216] Zizheng Guo, Runsheng Wang and **Yibo Lin***, "Provably Optimal Planar Pareto Nearest Neighbor Search with Double Monotone Chains", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C215] Jianqing Chen, Zhenkun Lin, Xun Jiang, Genggeng Liu*, **Yibo Lin** and Gang Du, "DSR: A Systematic Approach for Efficient Double-Sided Signal Routing", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C214] Siting Liu, Peng Xu, Peiyu Liao, Keren Zhu, **Yibo Lin** and Bei Yu*, "Submodular Maximization-Inspired Adaptive Routing Bend Space Planning", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C213] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "Efficient Warpage Simulation Of Complex 2.5-D/3-D IC Structures with Novel Meshing Algorithm and Layerwise Plate Theory", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

[C212] Yuhao Ji, Zizheng Guo, **Yibo Lin** and Bei Yu*, "HeteroSTA3D: A CPU-GPU Heterogeneous Static Timing Analysis Engine for 3D ICs", China Semiconductor Technology International Conference (CSTIC), Shanghai, Mar, 2026. (**Invited Paper**)

[C211] Jiarui Wang, Runzhe Tao, Jing Mai, Xun Jiang, Shenghua Wang, Cuiliu Yang, Haoyu Jie, Kan Huang, Richard Y. Sun and **Yibo Lin***, "TDM Signal Grouping and Package Pin Assignment for 2.5D Multi-FPGA Systems with Lookahead Placement", International Symposium on Field-Programmable Gate Arrays (FPGA), Monterey, CA, Feb 22-24, 2026.

[C210] Zizheng Guo, Haichuan Liu, Xizhe Shi, Shenglu Hua, Zuodong Zhang, Chunyuan Zhao, Runsheng Wang and **Yibo Lin***, "HeteroSTA: A CPU-GPU Heterogeneous Static Timing Analysis Engine with Holistic Industrial Design Support", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026. (**Invited Paper**)

[C209] Xizhe Shi, Zizheng Guo, **Yibo Lin***, Zuodong Zhang, Yun Liang and Runsheng Wang*, "HeteroLatch: A CPU-GPU Heterogeneous Latch-Aware Timing Analysis Engine", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C208] Kairong Guo, Haoran Lu, Rui Guo, Jiarui Wang, Chunyuan Zhao, Heng Wu, Runsheng Wang and **Yibo Lin***, "Standard Cell Layout Synthesis for Dual-Sided 3D-Stacked Transistors", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C207] Weijian Fan, Haoyi Zhang, Weibin Lin, Runsheng Wang and **Yibo Lin***, "MOSTAR: Multi-Stage Hierarchical Bayesian Optimization for Substructure-Aware High-Dimensional Analog Circuit Sizing", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C206] Chenhao Xue, Kezhi Li, Jiaxing Zhang, Yi Ren, Zhengyuan Shi, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun*, "AC-Refiner: Efficient Arithmetic Circuit Optimization Using Conditional Diffusion Models", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[C205] Jingchao Hu, **Yibo Lin**, Hao Yu, Quan Chen, Zhou Jin* and Cheng Zhuo, "IR Drop-Aware ECO: A Fast Approach to Minimize Layout and Timing Disturbance", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.

[J204] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu* and Martin Wong, "Sign-off Timing Considerations via Concurrent Routing Topology Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J203] Zhenkun Lin, Genggeng Liu*, Xing Huang, **Yibo Lin**, Jixin Zhang, Wenhao Liu and Ting-Chi Wang, "A Unified Deep Reinforcement Learning Approach for Constructing Rectilinear and Octilinear Steiner Minimum Tree", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J202] Jingchen Zhu, Chenhao Xue, Yiqi Chen, Zhao Wang, Chen Zhang, Yu Shen, Yifan Chen, Zekang Cheng, Yu Jiang, Tianqi Wang, **Yibo Lin**, Wei Hu, Bin Cui, Runsheng Wang, Yun Liang and Guangyu Sun*, "Theseus: Exploring Efficient Wafer-Scale Chip Design for Large Language Models", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J201] Yajuan Su, Zixi Liu, **Yibo Lin**, Xiaojing Su, Yuqin Wang, Xin Hong, Yujie Jiang, Pengyu Ren and Yayi Wei, "A Post-Routing Layout Optimization Framework for Lithography Process Window Enlargement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[J200] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, Runsheng Wang and **Yibo Lin***, "LEGALM 2.0: A Versatile Augmented Lagrangian Method-Based Methodology for Mixed-Cell-Height Legalization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)

[C199] Chunyuan Zhao, Jiarui Wang, Xun Jiang, Jincheng Lou and **Yibo Lin***, "GTA: GPU-Accelerated Track Assignment with Lightweight Lookup Table for Conflict Detection", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025. (**Best Paper Nomination**)

[C198] Yufan Du, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "Differentiable Physical Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C197] Yuhao Ji, Yuntao Lu, Zuodong Zhang, Zizheng Guo, **Yibo Lin** and Bei Yu*, "DiffCCD: Differentiable Concurrent Clock and Data Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C196] Yuan Pu, Yuhao Ji, Siying Yu, Zuodong Zhang, Zizheng Guo, Zhuolun He, **Yibo Lin**, David Z. Pan and Bei Yu*, "GPU Acceleration for Versatile Buffer Insertion", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C195] Yi Ren, Baokang Peng, Chenhao Xue, Kairong Guo, Yukun Wang, Guoyao Cheng, **Yibo Lin**, Lining Zhang and Guangyu Sun*, "Orthrus: Dual-Loop Automated Framework for System-Technology Co-Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C194] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin***, "IncreGPUSTA: GPU-Accelerated Incremental Static Timing Analysis for Iterative Design Flows", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C193] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "High-Resolution Full-Chip Thermal Resistance Extraction of BEOL Interconnects in 3-D ICs Considering Detailed Via Connectivity", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[C192] Chenchen Zhao, Zhengyuan Shi, Xiangyu Wen, Chengjie Liu, Yi Liu, Yunhao Zhou, Yuxiang

Zhao, Hefei Feng, Yinan Zhu, Waa Gwok-Wan, Xin Cheng, Weiyu Chen, Yongqi Fu, Chujie Chen, Chenhao Xue, Ying Wang, **Yibo Lin**, Jun Yang, Ning Xu, Xi Wang and Qiang Xu*, "MMCircuitEval: A Comprehensive Multimodal Circuit-Focused Benchmark for Evaluating LLMs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.

[J191] Bingyang Liu, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, **Yibo Lin***, Runsheng Wang and Ru Huang, "LayoutCopilot: An LLM-powered Multi-agent Collaborative Framework for Interactive Analog Layout Design", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2025.

[C190] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, Qiang Xu, Runsheng Wang and **Yibo Lin***, "DeepLayout: Learning Neural Representations of Circuit Placement Layout", International Conference on Machine Learning (ICML), Vancouver, Canada, Jul 13-19, 2025.

[C189] Lijie Zeng, Jiatai Sun, Xiao Wu, Dan Niu, Tianshi Wang, **Yibo Lin**, Zuochang Ye and Zhou Jin*, "G-SpNN: GPU-Accelerated Passivity Enforcement for S-Parameter Modeling with Neural Networks", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[C188] Ziyang Yu, Peng Xu, Zixiao Wang, Binwu Zhu, Qipan Wang, **Yibo Lin**, Runsheng Wang, Bei Yu* and Martin Wong, "SDM-PEB: Spatial-Depthwise Mamba for Enhanced Post-Exposure Bake Simulation", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[C187] Zizheng Guo, Yanqing Zhang, Runsheng Wang, **Yibo Lin** and Haoxing Ren, "GEM: GPU-Accelerated Emulator-Inspired RTL Simulation", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025. (**Best Paper Nomination**)

[C186] Xun Jiang, Haoran Lu, Yuxuan Zhao, Jiarui Wang, Zizheng Guo, Heng Wu, Bei Yu, Sung Kyu Lim, Runsheng Wang, Ru Huang and **Yibo Lin***, "A Systematic Approach for Multi-Objective Double-Side Clock Tree Synthesis", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[C185] Jiarui Wang, Yanjing Liu and **Yibo Lin***, "Synergistic Die-Level Router for Multi-FPGA System with Time-Division Multiplexing Optimization", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[C184] Yifan Chen, Jing Mai, Zuodong Zhang and **Yibo Lin***, "RUPlace: Optimizing Routability via Unified Placement and Routing Formulation", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.

[J183] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, **Yibo Lin***, Runsheng Wang and Ru Huang, "PDNNet: PDN-Aware GNN-CNN Heterogeneous Network for Dynamic IR Drop Prediction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2025.

[J182] Rui Guo, Haoran Lu, Jiacheng Sun, Xun Jiang, Lining Zhangm, Ming Li, **Yibo Lin**, Runsheng Wang, Heng Wu* and Ru Huang, "Design Optimization of Flip FET Standard Cells With Dual-Sided Pins for Ultimate Scaling", IEEE Transactions on Electron Devices (TED), Jun, 2025.

[C181] Haoyi Zhang, Shizhao Sun, **Yibo Lin**, Runsheng Wang and Jiang Bian, "AnalogXpert:

Automating Analog Topology Synthesis by Incorporating Circuit Design Expertise into Large Language Models", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.

[C180] Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, Seunggeun Kim, David Z. Pan, Runsheng Wang and **Yibo Lin***, "Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)

[C179] Kairong Guo and **Yibo Lin***, "Multi-Row Standard Cell Layout Synthesis with Enhanced Scalability", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.

[C178] Qipan Wang, **Yibo Lin***, Runsheng Wang and Ru Huang, "ATSim3.5D: A Multiscale Thermal Simulator for 3.5D-IC Systems based on Nonlinear Multigrid Method", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)

[C177] Chenhao Xue, Yi Ren, Jinwei Zhou, Kezhi Li, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun*, "DOMAC: Differentiable Optimization for High-Speed Multipliers and Multiply-Accumulators", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.

[C176] Yi Ren, Chenhao Xue, Jiaxing Zhang, Chen Zhang, Qiang Xu, **Yibo Lin** and Guangyu Sun*, "DiffuSE: Cross-Layer Design Space Exploration of DNN Accelerator via Diffusion-Driven Optimization", IEEE/ACM International Symposium of EDA (ISEDA), Hong Kong, May 9-12, 2025.

[C175] Haoran Lu, Xun Jiang, Yanbang Chu, Ziqiao Xu, Rui Guo, Wanyue Peng, **Yibo Lin**, Runsheng Wang, Heng Wu* and Ru Huang, "A Tale of Two Sides of Wafer: Physical Implementation and Block-Level PPA on Flip FET with Dual-sided Signals", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.

[C174] Tianxiang Zhu, Qipan Wang, **Yibo Lin***, Runsheng Wang and Ru Huang, "MORE-Stress: Model Order Reduction based Efficient Numerical Algorithm for Thermal Stress Simulation of TSV Arrays in 2.5D/3D IC", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.

[C173] Xizhe Shi, Zizheng Guo, **Yibo Lin***, Runsheng Wang and Ru Huang, "Handling Latch Loops in Timing Analysis with Improved Complexity and Divergent Loop Detection", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.

[C172] Haikang Diao, Haoyi Zhang, Jiahao Song, Haoyang Luo, **Yibo Lin**, Runsheng Wang, Yuan Wang and Xiyuan Tang*, "SEGA-DCIM: Design Space Exploration-Guided Automatic Digital CIM Compiler with Multiple Precision Support", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.

[C171] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, **Yibo Lin***, Runsheng Wang and Ru Huang, "LEGALM: Efficient Legalization for Mixed-Cell-Height Circuits with Linearized Augmented Lagrangian Method", ACM International Symposium on Physical Design (ISPD), Austin, TX, Mar 16-19, 2025.

[J170] Xun Jiang, Jiarui Wang, Jing Mai, Zhixiong Di and **Yibo Lin***, "A Robust FPGA Router

With Optimization of High-Fanout Nets and Intra-CLB Connections", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Mar, 2025.

[J169] Yuxuan Zhao, Peiyu Liao, Siting Liu, Jiaxi Jiang, **Yibo Lin** and Bei Yu*, "Analytical Heterogeneous Die-to-Die 3D Placement With Macros", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2025.

[C168] Che Chang, Boyang Zhang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, **Yibo Lin** and Tsung-Wei Huang*, "PathGen: An Efficient Parallel Critical Path Generation Algorithm", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan, 2025. (**Best Paper Nomination**)

[C167] Boyang Zhang, Che Chang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, Wan Luan Lee, Zizheng Guo, **Yibo Lin** and Tsung-Wei Huang*, "iTAP: An Incremental Task Graph Partitioner for Task-parallel Static Timing Analysis", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan, 2025.

[C166] Zizheng Guo, **Yibo Lin***, Runsheng Wang and Ru Huang, "Analyzing Timing in Shorter Time: A Journey through Heterogeneous Parallelism for Static Timing Analysis", IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Zhuhai, China, Oct 22-25, 2024. (**Invited Paper**)

[J165] Wenshuo Yue, Teng Zhang, Zhaokun Jing, Kai Wu, Yuxiang Yang, Zhen Yang, Yongqin Wu, Weihai Bu, Kai Zheng, Jin Kang, **Yibo Lin**, Yaoyu Tao, Bonan Yan*, Ru Huang and Yuchao Yang*, "A scalable universal Ising machine based on interaction-centric storage and compute-in-memory", Nature Electronics, Oct, 2024.

[C164] Qipan Wang, Xueqing Li, Tianyu Jia, **Yibo Lin***, Runsheng Wang and Ru Huang, "AT-Place2.5D: Analytical Thermal-Aware Chiplet Placement Framework for Large-Scale 2.5D-IC", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[C163] Chunyuan Zhao, Zizheng Guo, Rui Wang, Zaiwen Wen, Yun Liang and **Yibo Lin***, "HeLEM-GR: Heterogeneous Global Routing with Linearized Exponential Multiplier Method", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[C162] Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, Yufan Du, **Yibo Lin***, Runsheng Wang and Ru Huang, "HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[C161] Xiaohan Gao, Haoyi Zhang, Bingyang Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "Joint Placement Optimization for Hierarchical Analog/Mixed-Signal Circuits", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[C160] Yufan Du, Zizheng Guo, **Yibo Lin***, Runsheng Wang and Ru Huang, "Fusion of Global Placement and Gate Sizing with Differentiable Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024. (**Best Paper Nomination**)

[C159] Tianxiang Zhu, Qipan Wang, **Yibo Lin***, Runsheng Wang and Ru Huang, "FaStTherm: Fast and Stable Full-Chip Transient Thermal Predictor Considering Nonlinear Effects", IEEE/ACM

International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[C158] Jing Mai, Zuodong Zhang, **Yibo Lin***, Runsheng Wang and Ru Huang, "MORPH: More Robust ASIC Placement for Hybrid Region Constraint Management", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

[J157] Tsung-Yi Ho, Sadaf Khan, Jinwei Liu, Yi Liu, Zhengyuan Shi, Ziyi Wang, Qiang Xu*, Evangeline F.Y. Young, Bei Yu, Ziyang Zheng, Binwu Zhu, Keren Zhu, Yiqi Che, Yun Liang, **Yibo Lin**, Guojie Luo, Guangyu Sun, Runsheng Wang, Xinming Wei, Chenhao Xue, Haoyi Zhang, Zuodong Zhang, Yuxiang Zhao, Sunan Zou, Lei Chen, Yu Huang, Min Li, Dimitrios Tsaras, Mingxuan Yuan, Hui-Ling Zhen, Zhufei Chu, Wenji Fang, Xingquan Li and Zhiyao Xie, "Large Circuit Models: Opportunities and Challenges", Science China Information Sciences, Sep, 2024.

[C156] Jiarui Wang, Xun Jiang and **Yibo Lin***, "Top-Level Routing for Multiply-Instantiated Blocks with Topology Hashing", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C155] Yufan Du, Zizheng Guo, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin***, Runsheng Wang and Ru Huang, "PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C154] Haoyi Zhang, Jiahao Song, Xiaohan Gao, Xiyuan Tang, **Yibo Lin***, Runsheng Wang and Ru Huang, "EasyACIM: An End-to-End Automated Analog CIM with Synthesizable Architecture and Agile Design Space Exploration", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C153] Zichen Kong, Xiyuan Tang*, Wei Shi, Yiheng Du, **Yibo Lin** and Yuan Wang, "PVTSizing: A TuRBO-RL-Based Batch-Sampling Optimization Framework for PVT-Robust Analog Circuit Synthesis", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C152] Yuan Pu, Fangzhou Liu, Yu Zhang, Zhuolun He, Kai-Yuan Chao, **Yibo Lin** and Bei Yu*, "Lesyn: Placement-aware Logic Resynthesis for Non-Integer Multiple-Cell-Height Designs", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C151] Wan Luan Lee, Dian-Lun Lin, Tsung-Wei Huang*, Shui Jiang, Tsung-Yi Ho, **Yibo Lin** and Bei Yu, "G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C150] Chenhao Xue, Chen Zhang, Xun Jiang, Gao Zhutianya, **Yibo Lin** and Guangyu Sun*, "Oltron: Algorithm-Hardware Co-design for Outlier-Aware Quantization of LLMs with Inter-/Intra-Layer Adaptation", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.

[C149] Haoran Lu, Yandong Ge, Xun Jiang, Jiacheng Sun, Wanyue Peng, Rui Guo, Ming Li, **Yibo Lin**, Runsheng Wang, Heng Wu* and Ru Huang, "First Experimental Demonstration of Self-Aligned Flip FET (FFET): A Breakthrough Stacked Transistor Technology with 2.5T Design, Dual-Side Active and Interconnects", IEEE Symposium on VLSI Technology and Circuits (VLSI), Honolulu, HI, Jun 16-20, 2024.

[J148] Peiyu Liao, Yuxuan Zhao, Dawei Guo, **Yibo Lin** and Bei Yu*, "Analytical Die-to-Die 3D

Placement With Bistratal Wirelength Model and GPU Acceleration", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2024.

[C147] Xiaohan Gao, Haoyi Zhang, Zhu Pan, **Yibo Lin***, Runsheng Wang and Ru Huang, "Migrating Standard Cells for Multiple Drive Strengths by Routing Imitation", IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024.

[C146] Qipan Wang, Tianxiang Zhu, **Yibo Lin***, Runsheng Wang and Ru Huang, "ATSim3D: Towards Accurate Thermal Simulator for Heterogeneous 3D IC Systems Considering Nonlinear Leakage and Conductivity", IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024. (**Honorable Mention Paper Award**)

[C145] Jing Mai, Jiarui Wang, Yifan Chen, Zizheng Guo, Xun Jiang, Yun Liang and **Yibo Lin***, "OpenPARF 3.0: Robust Multi-Electrostatics Based FPGA Macro Placement Considering Cascaded Macros Groups and Fence Regions", IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024. (**Best Paper Award**)

[C144] Xun Jiang, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin***, Runsheng Wang and Ru Huang, "CircuitNet 2.0: An Advanced Dataset for Promoting Machine Learning Innovations in Realistic Chip Design Environment", International Conference on Learning Representations (ICLR), Vienna, Austria, May 7-11, 2024.

[J143] Yufei Chen, Zizheng Guo, Runsheng Wang, Ru Huang, **Yibo Lin** and Cheng Zhuo, "Dynamic Supply Noise Aware Timing Analysis With JIT Machine Learning Integration", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), May, 2024. (accepted)

[C142] Zizheng Guo, Tsung-Wei Huang, Zhou Jin, Cheng Zhuo, **Yibo Lin***, Runsheng Wang and Ru Huang, "Heterogeneous Static Timing Analysis with Advanced Delay Calculator", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Valencia, Spain, Mar 24-28, 2024.

[C141] Haoyi Zhang, Xiaohan Gao, Zilong Shen, Jiahao Song, Xiaoxu Cheng, Xiyuan Tang, **Yibo Lin***, Runsheng Wang and Ru Huang, "SAGERoute 2.0: Hierarchical Analog and Mixed Signal Routing Considering Versatile Routing Scenarios", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Valencia, Spain, Mar 24-28, 2024.

[C140] Yuan Pu, Tinghuan Chen, Zhuolun He, Chen Bai, Haisheng Zheng, **Yibo Lin** and Bei Yu*, "IncreMacro: Incremental Macro Placement Refinement", ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024. (**Best Paper Nomination**)

[C139] Yu Zhang, Yuan Pu, Fangzhou Liu, Peiyu Liao, Kaiyuan Chao, Keren Zhu, **Yibo Lin** and Bei Yu*, "Multi-Electrostatics Based Placement for Non-Integer Multiple-Height Cells", ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024.

[C138] Siting Liu, Jiaxi Jiang, Zhuolun He, Ziyi Wang, **Yibo Lin** and Bei Yu*, "Routing-aware Legal Hybrid Bonding Terminal Assignment for 3D Face-to-Face Stacked ICs", ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024.

[J137] Zhixiong Di*, Runzhe Tao, Jing Mai, Lin Chen and **Yibo Lin**, "LEAPS: Topological-Layout-Adaptable Multi-Die FPGA Placement for Super Long Line Minimization", IEEE Transactions on Circuits and Systems I, Mar, 2024.

[J136] Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin***, "Multielectrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2024.

[C135] Cheng-Hsiang Chiu, Zhicheng Xiong, Zizheng Guo, Tsung-Wei Huang* and **Yibo Lin**, "An Efficient Task-parallel Pipeline Programming Framework", International Conference on High-Performance Computing in Asia-Pacific Region (HPC Asia), Nagoya, Japan, Jan, 2024.

[J134] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2023.

[J133] Zhuomin Chai, Yuxiang Zhao, Wei Liu*, **Yibo Lin***, Runsheng Wang and Ru Huang, "CircuitNet: An Open-Source Dataset for Machine Learning in VLSI CAD Applications with Improved Domain-Specific Evaluation Metric and Learning Strategies", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2023.

[J132] Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin***, "OpenPARF: 基于深度学习工具包的大规模异构 FPGA 开源布局布线框架", 电子与信息学报, 2023.

[J131] Zuodong Zhang, Zizheng Guo, **Yibo Lin***, Meng Li*, Runsheng Wang and Ru Huang, "AVATAR: An Aging- and Variation-Aware Dynamic Timing Analyzer for Error-Efficient Computing", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2023.

[J130] Guannan Guo, Tsung-Wei Huang*, **Yibo Lin**, Zizheng Guo, Sushma Yellapragada and Martin Wong, "A GPU-accelerated Framework for Path-based Timing Analysis", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2023.

[C129] Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin***, "Stronger Mixed-Size Placement Backbone Considering Second-Order Information", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.

[J128] Peiyu Liao, Dawei Guo, Zizheng Guo, Siting Liu, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "DREAMPlace 4.0: Timing-driven Placement with Momentum-based Net Weighting and Lagrangian-based Refinement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2023.

[J127] Xiaohan Gao, Haoyi Zhang, Siyuan Ye, Mingjie Liu, David Z. Pan, Linxiao Shen, Runsheng Wang, **Yibo Lin*** and Ru Huang, "Post-Layout Simulation Driven Analog Circuit Sizing", SCIENCE CHINA Information Sciences, Oct, 2023.

[C126] Jing Mai, Jiaru Wang, Zhixiong Di, Guojie Luo, Yun Liang and **Yibo Lin***, "OpenPARF: An Open-Source Placement and Routing Framework for Large-Scale Heterogeneous FPGAs with Deep Learning Toolkit", International Conference on ASIC (ASICON), Nanjing, China, Oct, 2023. (**Invited Paper**)

[C125] Xun Jiang, Zizheng Guo, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin***, Runsheng Wang and Ru Huang, "Accelerating Routability and Timing Optimization with Open-Source AI4EDA Dataset CircuitNet and Heterogeneous Platforms", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct, 2023. (**Invited Paper**)

[C124] Kexing Zhou, Yun Liang*, **Yibo Lin**, Runsheng Wang and Ru Huang, "Khronos: Fusing Memory Access for Improved Hardware RTL Simulation", IEEE/ACM International Symposium on Microarchitecture (MICRO), Toronto, Canada, Oct, 2023.

[J123] Binwu Zhu, Xinyun Zhang, **Yibo Lin**, Bei Yu and Martin Wong, "DRC-SG 2.0: Efficient Design Rule Checking Script Generation via Key Information Extraction", ACM Transactions on Design Automation of Electronic Systems (TODAES), Sep, 2023.

[C122] Zizheng Guo, Zuodong Zhang, Xun Jiang, Wuxi Li, **Yibo Lin***, Runsheng Wang and Ru Huang, "General-Purpose Gate-Level Simulation with Partition-Agnostic Parallelism", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[C121] Qipan Wang, Ping Liu, Ligguo Jiang, Mingjie Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "MTL-Designer: An Integrated Flow for Analysis and Synthesis of Microstrip Transmission Line", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[C120] Peiyu Liao, Hongduo Liu, **Yibo Lin***, Bei Yu* and Martin Wong, "On a Moreau Envelope Wirelength Model for Analytical Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[C119] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu and Martin Wong*, "Concurrent Sign-off Timing Optimization via Deep Steiner Points Refinement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[C118] Su Zheng, Lancheng Zou, Siting Liu, **Yibo Lin**, Bei Yu and Martin Wong*, "Mitigating Distribution Shift for Congestion Optimization in Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[C117] Yu Zhang, Yifan Chen, Zhonglin Xie, Hong Xu, Zaiwen Wen, **Yibo Lin*** and Bei Yu*, "LRSDP: Low-Rank SDP for Triple Patterning Lithography Layout Decomposition", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.

[J116] Siting Liu, Yuan Pu, Peiyu Liao, Hongzhong Wu, Rui Zhang, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2023.

[C115] Yuxiang Zhao, Zhuomin Chai, **Yibo Lin***, Runsheng Wang and Ru Huang, "HybridNet: Dual-Branch Fusion of Geometrical and Topological Views for VLSI Congestion Prediction", IEEE/ACM International Symposium of EDA (ISEDA), Nanjing, China, May 8-11, 2023.

[C114] Haoyi Zhang, Xiaohan Gao, **Yibo Lin***, Runsheng Wang and Ru Huang, "Multi-Scenario Analog and Mixed-Signal Circuit Routing with Agile Human Interaction", IEEE/ACM International Symposium of EDA (ISEDA), Nanjing, China, May 8-11, 2023.

[C113] Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)

[C112] Zuodong Zhang, Meng Li*, **Yibo Lin**, Runsheng Wang and Ru Huang, "READ: Reliability-Enhanced Accelerator Dataflow Optimization using Critical Input Pattern Reduction", IEEE/ACM

Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023.

[J111] Xiaohan Gao, Haoyi Zhang, Mingjie Liu, Linxiao Shen, David Z. Pan, **Yibo Lin***, Runsheng Wang and Ru Huang, "Interactive Analog Layout Editing with Instant Placement and Routing Legalization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Mar, 2023.

[J110] **Yibo Lin**, Avi Ziv and Haoxing Ren, "Introduction to the Special Issue on Machine Learning for CAD/EDA", ACM Transactions on Design Automation of Electronic Systems (TODAES), Mar, 2023.

[J109] Xinfa Zhang, Zuodong Zhang, **Yibo Lin***, Zhigang Ji*, Runsheng Wang and Ru Huang, "Efficient Aging-Aware Standard Cell Library Characterization Based on Sensitivity Analysis", IEEE Transactions on Circuits and Systems II: Express Briefs, Feb, 2023.

[C108] Yifan Chen, Jing Mai, Xiaohan Gao, Muhan Zhang and **Yibo Lin***, "MacroRank: Ranking Macro Placement Solutions Leveraging Translation Equivariancy", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 16-19, 2023.

[C107] Jiarui Wang, Jing Mai, Zhixiong Di and **Yibo Lin***, "A Robust FPGA Router with Concurrent Intra-CLB Rerouting", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 16-19, 2023.

[J106] Junchi Yan*, Xianglong Lyu, Ruoyu Cheng and **Yibo Lin**, "Towards Machine Learning for Placement and Routing in Chip Design: a Methodological Overview", arXiv preprint, 2022.

[J105] Wei Li, Jialu Xia, Yuzhe Ma, Jialu Li, **Yibo Lin** and Bei Yu*, "Adaptive Layout Decomposition with Graph Embedding Neural Networks", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2022.

[C104] Zizheng Guo, Feng Gu and **Yibo Lin***, "GPU-Accelerated Rectilinear Steiner Tree Generation", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022.

[C103] Qipan Wang, Xiaohan Gao, **Yibo Lin***, Runsheng Wang and Ru Huang, "DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)

[C102] **Yibo Lin***, Xiaohan Gao, Haoyi Zhang, Runsheng Wang and Ru Huang, "Intelligent and Interactive Analog Layout Design Automation", IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Nanjing, China, Oct 25-28, 2022. (**Invited Paper**)

[J101] Zizheng Guo, Mingwei Yang, Tsung-Wei Huang and **Yibo Lin***, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2022.

[J100] Martin Rapp, Hussam Amrouch, **Yibo Lin**, Bei Yu, David Z. Pan, Marilyn Wolf and Jörg Henkel*, "MLCAD: A Survey of Research in Machine Learning for CAD", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2022. (**Invited Keynote paper**)

[C99] Binwu Zhu, Xinyun Zhang, **Yibo Lin**, Bei Yu* and Martin Wong, "Efficient Design Rule Checking Script Generation via Key Information Extraction", ACM/IEEE Workshop on Machine Learning for CAD (MLCAD), Snowbird, Utah, Sep 12-13, 2022.

[J98] **Yibo Lin**, Tong Qu, Zongqing Lu, Yajuan Su* and Yayi Wei*, "Asynchronous Reinforcement Learning Framework and Knowledge Transfer for Net Order Exploration in Detailed Routing", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2022.

[J97] Zhuomin Chai, Yuxiang Zhao, **Yibo Lin***, Wei Liu, Runsheng Wang and Ru Huang, "CircuitNet: An Open-Source Dataset for Machine Learning Applications in Electronic Design Automation (EDA)", SCIENCE CHINA Information Sciences, Sep, 2022.

[C96] Jing Mai, Yibai Meng, Zhixiong Di and **Yibo Lin***, "Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity and Clock Feasibility", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

[C95] Zizheng Guo and **Yibo Lin***, "Differentiable-Timing-Driven Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

[C94] Zizheng Guo, Mingjie Liu, Jiaqi Gu, Shuhan Zhang, David Z. Pan and **Yibo Lin***, "A Timing Engine Inspired Graph Neural Network Model for Pre-Routing Slack Prediction", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

[C93] Zuodong Zhang, Zizheng Guo, **Yibo Lin***, Runsheng Wang and Ru Huang, "AVATAR: An Aging- and Variation-Aware Dynamic Timing Analyzer for Application-based DVAFS", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

[C92] Bowen Wang, Guibao Shen, Dong Li, Jianye Hao, Wulong Liu, Yu Huang, Hongzhong Wu, **Yibo Lin**, Guangyong Chen and Pheng Ann Heng, "LHNN: Lattice Hypergraph Neural Network for VLSI Congestion Prediction", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

[J91] Tsung-Wei Huang*, Dian-Lun Lin, Chun-Xun Lin and **Yibo Lin**, "Taskflow: A Lightweight Parallel and Heterogeneous Task Graph Computing System", IEEE Transactions on Parallel and Distributed Systems (TPDS), Jun, 2022.

[C90] Zuodong Zhang, Zizheng Guo, **Yibo Lin***, Runsheng Wang and Ru Huang, "EventTimer: Fast and Accurate Event-Based Dynamic Timing Analysis", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.

[C89] Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)

[C88] Peiyu Liao, Siting Liu, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "DREAMPlace 4.0: Timing-driven Global Placement with Momentum-based Net Weighting", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.

[C87] Haoyu Yang, Kit Fung, Yuxuan Zhao, **Yibo Lin** and Bei Yu*, "Mixed-Cell-Height Legalization on CPU-GPU Heterogeneous Systems", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.

[C86] Xun Jiang, **Yibo Lin** and Zhongfeng Wang*, "FPGA-Accelerated Maze Routing Kernel for VLSI Designs", IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), Virtual Conference, Jan 17-20, 2022.

[C85] Kexing Zhou, Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "Efficient Critical Paths Search Algorithm using Mergeable Heap", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Virtual Conference, Jan 17-20, 2022.

[J84] Yibai Meng, Wuxi Li, **Yibo Lin*** and David Z. Pan, "elfPlace: Electrostatics-based Placement for Large-Scale Heterogeneous FPGAs", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jan, 2022.

[C83] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

[C82] Guannan Guo, Tsung-Wei Huang*, **Yibo Lin** and Martin Wong, "GPU-accelerated Path-based Timing Analysis", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

[C81] Zizheng Guo, Jing Mai and **Yibo Lin***, "Ultrafast CPU/GPU Kernels for Density Accumulation in Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

[C80] Xiaohan Gao, Mingjie Liu, David Z. Pan and **Yibo Lin***, "Interactive Analog Layout Editing with Instant Placement Legalization", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.

[J79] **Yibo Lin***, "模拟电路版图自动化与智能设计", 中国计算机学会通讯, Dec, 2021.

[J78] Wei Li, Yuzhe Ma, Qi Sun, Zhang Lu, **Yibo Lin**, Iris Hui-Ru Jiang, Bei Yu* and David Z. Pan, "OpenMPL: An Open Source Layout Decomposer", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2021.

[C77] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "HeteroCPPR: Accelerating Common Path Pessimism Removal with Heterogeneous CPU-GPU Parallelism", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Virtual Conference, Nov 01-04, 2021.

[C76] Guannan Guo, Tsung-Wei Huang*, **Yibo Lin** and Martin Wong, "GPU-accelerated Critical Path Generation with Path Constraints", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Virtual Conference, Nov 01-04, 2021.

[C75] Tong Qu, **Yibo Lin**, Tianyang Gai, Xiaojing Su, Shuhan Wang, Bojie Ma, Yajuan Su* and Yayi Wei*, "Litho-Aware Redundant Local-Loop Insertion Framework With Convolutional Neural Network", Proceedings of SPIE, San Jose, CA, Sep 27, 2021.

[J74] Tsung-Wei Huang*, **Yibo Lin**, Chun-Xun Lin, Guannan Guo and Martin Wong, "Cpp-Taskflow: A General-purpose Parallel Task Programming System at Scale", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2021.

[J73] **Yibo Lin***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, "DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

[J72] Junzhe Cai, Changhao Yan*, Yudong Tao, **Yibo Lin**, Sheng-Guo Wang, David Z. Pan and Xuan Zeng, "A Novel and Unified Full-chip CMP Model Aware Dummy Fill Insertion Framework with SQP-Based Optimization Method", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Mar, 2021.

[J71] Mohamed Baker Alawieh, **Yibo Lin**, Zaiwei Zhang, Meng Li, Qixing Huang and David Z. Pan*, "GAN-SRAF: Sub-Resolution Assist Feature Generation using Generative Adversarial Networks", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2021.

[J70] **Yibo Lin***, Xiaohan Gao, Tinghuan Chen and Bei Yu*, "机器学习辅助数字集成电路后端设计方法", 微纳电子与智能制造, Feb, 2021.

[J69] Cheng Zhuo*, Zizheng Guo, Xiao Dong, Qing He and **Yibo Lin**, "先进工艺下的数字签核", 微纳电子与智能制造, Feb, 2021.

[C68] Tong Qu, **Yibo Lin**, Zongqing Lu, Yajuan Su* and Yayi Wei*, "Asynchronous Reinforcement Learning Framework for Net Order Exploration in Detailed Routing", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.

[C67] Siting Liu, Qi Sun, Peiyu Liao, **Yibo Lin** and Bei Yu*, "Global Placement with Deep Learning-Enabled Explicit Routability Optimization", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.

[C66] Hongjia Li, Mengshu Sun, Tianyun Zhang, Olivia Chen, Nobuyuki Yoshikawa, Bei Yu, Yanzhi Wang* and **Yibo Lin**, "Towards AQFP-Capable Physical Design Automation", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.

[C65] Xiaohan Gao, Chenhui Deng, Mingjie Liu, Zhiru Zhang, David Z. Pan and **Yibo Lin***, "Layout Symmetry Annotation for Analog Circuits with Graph Neural Networks", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 18-21, 2021.

[C64] **Yibo Lin***, "Deep Learning for Mask Synthesis and Verification: A Survey", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 18-21, 2021. (**Invited Paper**)

[J63] **Yibo Lin***, Wuxi Li, Jiaqi Gu, Haoxing Ren, Brucek Khailany and David Z. Pan, "ABCD-Place: Accelerated Batch-based Concurrent Detailed Placement on Multi-threaded CPUs and GPUs", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2020.

[C62] Jiaqi Gu, Zixuan Jiang, **Yibo Lin** and David Z. Pan*, "DREAMPlace 3.0: Multi-Electrostatics Based Robust VLSI Placement with Region Constraints", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020.

[C61] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "GPU-Accelerated Static Timing Analysis", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020.

[C60] **Yibo Lin***, "GPU Acceleration in VLSI Back-end Design: Overview and Case Studies", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020. (**Invited tutorial**)

[C59] Wei Ye, Mohamed Baker Alawieh, Yuki Watanabe, Shigeki Nojima, **Yibo Lin** and David Z. Pan*, "TEMPO: Fast Mask Topography Effect Modeling with Deep Learning", ACM International Symposium on Physical Design (ISPD), Taipei, Taiwan, Sep 20-23, 2020. (**Best Paper Award**)

[J58] Jing Chen, Mohamed Baker Alawieh, **Yibo Lin**, Maolin Zhang, Jun Zhang, Yufeng Guo* and David Z. Pan, "Automatic Selection of Structure Parameters of Silicon on Insulator Lateral Power Device Using Bayesian Optimization", IEEE Electron Device Letters (EDL), Sep, 2020.

[C57] Wei Li, Jialu Xia, Yuzhe Ma, Jialu Li, **Yibo Lin** and Bei Yu*, "Adaptive Layout Decomposition with Graph Embedding Neural Networks", ACM/IEEE Design Automation Conference (DAC), San Francisco, Jul 19-23, 2020.

[J56] Ying Chen, **Yibo Lin**, Rui Chen, Lisong Dong, Ruixuan Wu, Tianyang Gai, Le Ma, Yajuan Su* and Yayi Wei*, "EUV Multilayer Defect Characterization via Cycle-Consistent Learning", Optics Express, Jun, 2020.

[C55] **Yibo Lin***, David Z. Pan, Haoxing Ren and Brucek Khailany, "DREAMPlace 2.0: Open-Source GPU-Accelerated Global and Detailed Placement for Large-Scale VLSI Designs", China Semiconductor Technology International Conference (CSTIC), Shanghai, China, Jun, 2020. (**Invited Paper**)

[J54] Jing Chen, Mohamed Baker Alawieh, **Yibo Lin**, Maolin Zhang, Jun Zhang, Yufeng Guo* and David Z. Pan, "Powernet: SOI Lateral Power Device Breakdown Prediction With Deep Neural Networks", IEEE Access, Feb, 2020.

[J53] Ying Chen, **Yibo Lin**, Lisong Dong, Tianyang Gai, Rui Chen, Yajuan Su*, Yayi Wei* and David Z. Pan, "SoulNet: Ultrafast Optical Source Optimization Utilizing Generative Neural Networks for Advanced Lithography", Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Nov, 2019.

========== Below are publications during Ph.D. and Postdoc ==========

Book Chapters

[B1] **Yibo Lin** and David Z. Pan, "Machine Learning in Physical Verification, Mask Synthesis, and Physical Design", Machine Learning in VLSI Computer-Aided Design, Springer, 2018, edited by Abe Elfedel, Duane Boning and Xin Li. (**Invited Book Chapter**)

Conference and Journal Papers (* denotes corresponding authors)

Summary: DAC (7), ICCAD (7), IEEE TCAD (10), DATE (1), etc.

[J52] Hao Chen, Mingjie Liu, Biying Xu, Keren Zhu, Xiyuan Tang, Shaolan Li, **Yibo Lin**, Nan Sun and David Z. Pan*, "MAGICAL: An Open-Source Fully Automated Analog IC Layout System from Netlist to GDSII", IEEE Design & Test, Apr, 2021.

[C51] Rachel Selina Rajarathnam, **Yibo Lin**, Yier Jin and David Z. Pan*, "ReGDS: A Reverse Engineering Framework from GDSII to Gate-level Netlist", IEEE International Workshop on Hardware-Oriented Security and Trust (HOST), San Jose, CA, May 4, 2020.

[C50] Mingjie Liu, Wuxi Li, Keren Zhu, Biying Xu, **Yibo Lin**, Linxiao Shen, Xiyuan Tang, Nan Sun and David Z. Pan*, "S3DET: Detecting System Symmetry Constraints for Analog Circuits with Graph Similarity", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jan 13-16, 2020. (**Best Paper Nomination**)

[C49] Mohamed Baker Alawieh, Wuxi Li, **Yibo Lin**, Love Singhal, Mahesh Iyer and David Z. Pan*, "High-Definition Routing Congestion Prediction for Large-Scale FPGAs", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jan 13-16, 2020.

[C48] Wuxi Li, **Yibo Lin** and David Z. Pan*, "elfPlace: Electrostatics-based Placement for Large-Scale Heterogeneous FPGAs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.

[C47] Keren Zhu, Mingjie Liu, **Yibo Lin**, Biying Xu, Shaolan Li, Xiyuan Tang, Nan Sun and David Z. Pan*, "GeniusRoute: A New Analog Routing Paradigm Using Generative Neural Network Guidance", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.

[C46] Chengyue Gong, Zixuan Jiang, Dilin Wang, **Yibo Lin**, Qiang Liu and David Z. Pan*, "Mixed Precision Neural Architecture Search for Energy Efficient Deep Learning", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.

[C45] Biying Xu, Keren Zhu, Mingjie Liu, **Yibo Lin**, Shaolan Li, Xiyuan Tang, Nan Sun and David Z. Pan*, "MAGICAL: Toward Fully Automated Analog IC Layout Leveraging Human and Machine Intelligence", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019. (**Invited Paper**)

[J44] **Yibo Lin**, Meng Li, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima and David Z. Pan*, "Data Efficient Lithography Modeling with Transfer Learning and Active Data Selection", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2019.

[C43] Wei Li, Yuzhe Ma, Qi Sun, **Yibo Lin**, Iris Hui-Ru Jiang, Bei Yu and David Z Pan, "OpenMPL: An Open Source Layout Decomposer", International Conference on ASIC (ASICON), Chongqing, China, Oct, 2019. (**Invited Paper**)

[C42] **Yibo Lin**, Shounak Dhar, Wuxi Li, Haoxing Ren, Brucek Khailany and David Z. Pan*, "DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Award**)

[C41] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan*, "LithoGAN: End-to-End Lithography Modeling with Generative Adversarial Networks", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Nomination**)

[C40] Biying Xu, **Yibo Lin**, Xiyuan Tang, Shaolan Li, Linxiao Shen, Nan Sun and David Z. Pan*, "WellGAN: Generative-Adversarial-Network-Guided Well Generation for Analog/Mixed-Signal Circuit Layout", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.

[C39] Mohamed Baker Alawieh, **Yibo Lin**, Zaiwei Zhang, Meng Li, Qixing Huang and David Z. Pan*, "GAN-SRAF: Sub-Resolution Assist Feature Generation Using Conditional Generative Adversarial Networks", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.

[C38] **Yibo Lin**, Zhao Song and Lin F. Yang*, "Towards a Theoretical Understanding of Hashing-Based Neural Nets", International Conference on Artificial Intelligence and Statistics (AIS-

TATS), Okinawa, Japan, Apr 16-18, 2019.

[C37] Biying Xu, Shaolan Li, Chak-Wa Pui, Derong Liu, Linxiao Shen, **Yibo Lin**, Nan Sun and David Z. Pan*, "Device Layer-Aware Analytical Placement for Analog Circuits", ACM International Symposium on Physical Design (ISPD), San Francisco, CA, Apr 14-17, 2019. (**Best Paper Nomination**)

[J36] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su*, Yayi Wei* and David Z. Pan, "Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2019.

[C35] Wei Ye, Mohamed Baker Alawieh, Meng Li, **Yibo Lin** and David Z. Pan*, "Litho-GPA: Gaussian Process Assurance for Lithography Hotspot Detection", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Florence, Italy, Mar 25-29, 2019.

[J34] Jing Chen, **Yibo Lin**, Yufeng Guo, Maolin Zhang, Mohamed Baker Alawieh and David Z. Pan*, "Lithography Hotspot Detection Using a Double Inception Module Architecture", Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Mar, 2019.

[C33] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su*, Yayi Wei* and David Z. Pan, "Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.

[C32] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan*, "Tackling Signal Electromigration with Learning-Based Detection and Multistage Mitigation", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.

[C31] Wei Ye, **Yibo Lin**, Meng Li, Qiang Liu and David Z. Pan*, "LithoROC: Lithography Hotspot Detection with Explicit ROC Optimization", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019. (**Invited Paper**)

[C30] **Yibo Lin**, Mohamed Baker Alawieh, Wei Ye and David Z. Pan*, "Machine Learning for Yield Learning and Optimization", IEEE International Test Conference (ITC), Phoenix, Arizona, Oct, 2018. (**Invited Paper**)

[J29] **Yibo Lin**, Bei Yu, Meng Li and David Z. Pan*, "Layout Synthesis for Topological Quantum Circuits with 1D and 2D Architectures", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2018.

[J28] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z Pan, "A practical split manufacturing framework for trojan prevention via simultaneous wire lifting and cell insertion", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2018.

[C27] Jiong Zhang, **Yibo Lin**, Zhao Song and Inderjit S Dhillon*, "Learning Long Term Dependencies via Fourier Recurrent Units", International Conference on Machine Learning (ICML), Stockholm, Sweden, Jun 10-15, 2018.

[J26] Xiaoqing Xu, **Yibo Lin**, Meng Li, Tetsuaki Matsunawa, Shigeki Nojima, Chikaaki Kodama, Toshiya Kotani and David Z. Pan*, "Subresolution Assist Feature Generation With Supervised Data Learning", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.

[J25] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu,

Zhuo Li, Charles J Alpert and David Z. Pan*, "MrDP: Multiple-row detailed placement of heterogeneous-sized cells for advanced nodes", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.

[J24] Wuxi Li, **Yibo Lin**, Meng Li, Shounak Dhar and David Z. Pan*, "UTPlaceF 2.0: A High-Performance Clock-Aware FPGA Placement Engine", ACM Transactions on Design Automation of Electronic Systems (TODAES), Jun, 2018.

[C23] **Yibo Lin**, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima, Meng Li and David Z. Pan*, "Data Efficient Lithography Modeling with Residual Neural Networks and Transfer Learning", ACM International Symposium on Physical Design (ISPD), Monterey, CA, Mar 25-28, 2018.

[C22] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z. Pan*, "A Practical Split Manufacturing Framework for Trojan Prevention via Simultaneous Wire Lifting and Cell Insertion", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.

[C21] Che-Lun Hsu, Shaofeng Guo, **Yibo Lin**, Xiaoqing Xu, Meng Li, Runsheng Wang, Ru Huang and David Z Pan, "Layout-dependent aging mitigation for critical path timing", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.

[C20] **Yibo Lin**, Peter Debacker, Darko Trivkovic, Ryoung-han Kim, Praveen Raghavan and David Z. Pan*, "Patterning Aware Design Optimization of Selective Etching in N5 and Beyond", IEEE International Conference on Computer Design (ICCD), Boston, MA, Nov 5-8, 2017.

[C19] **Yibo Lin**, Xiaoqing Xu, Jiaojiao Ou and David Z Pan, "Machine learning for mask/wafer hotspot detection and mask synthesis", Photomask Technology, Oct 16, 2017. (**Invited Paper**)

[J18] **Yibo Lin**, Bei Yu and David Z. Pan*, "High performance dummy fill insertion with coupling and uniformity constraints", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2017.

[C17] Wei Ye, **Yibo Lin**, Xiaoqing Xu, Wuxi Li, Yiwei Fu, Yongsheng Sun, Canhui Zhan and David Z. Pan*, "Placement Mitigation Techniques for Power Grid Electromigration", IEEE International Symposium on Low Power Electronics and Design (ISLPED), Taipei, Jul 24-26, 2017.

[J16] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan*, "Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.

[J15] Xiaoqing Xu, **Yibo Lin**, Meng Li, Jiaojiao Ou, Brian Cline and David Z. Pan*, "Redundant local-Loop insertion for unidirectional routing", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.

[C14] Xiaoqing Xu, **Yibo Lin**, Vinicius Livramento and David Z. Pan*, "Concurrent Pin Access Optimization for Unidirectional Routing", ACM/IEEE Design Automation Conference (DAC), Austin, TX, Jun 18-22, 2017.

[J13] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan*, "Stitch aware detailed placement for multiple e-beam lithography", Integration, the VLSI Journal, Jun,

2017. (**Best Paper Award**)

[J12] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan*, "Triple/quadruple patterning layout decomposition via linear programming and iterative rounding", Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Jun, 2017.

[C11] Jiaojiao Ou, Bei Yu, Xiaoqing Xu, Joydeep Mitra, **Yibo Lin** and David Z. Pan*, "DSAR: DSA aware routing with simultaneous DSA guiding pattern and double patterning assignment", ACM International Symposium on Physical Design (ISPD), Portland, OR, Mar 19-22, 2017.

[C10] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu, Zhuo Li, Charles J Alpert and David Z. Pan*, "MrDP: Multiple-row detailed placement of heterogeneous-sized cells for advanced nodes", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.

[C9] Yudong Tao, Changhao Yan*, **Yibo Lin**, Sheng-Guo Wang, David Z. Pan and Xuan Zeng, "A novel unified dummy fill insertion framework with SQP-based optimization method", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.

[C8] **Yibo Lin**, Bei Yu and David Z. Pan*, "Detailed placement in advanced technology nodes: a survey", IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Hangzhou, China, Oct 25-28, 2016. (**Invited Paper**)

[J7] Bei Yu, Xiaoqing Xu, Subhendu Roy, **Yibo Lin**, Jiaojiao Ou and David Z. Pan*, "Design for manufacturability and reliability in extreme-scaling VLSI", Science China Information Sciences, May, 2016. (**Invited Paper**)

[C6] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan*, "Triple/quadruple patterning layout decomposition via novel linear programming and iterative rounding", Proceedings of SPIE, San Jose, CA, Feb 21-25, 2016. (**Best Student Paper Award**)

[C5] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan*, "Stitch aware detailed placement for multiple e-beam lithography", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Macau, China, Jan 25-28, 2016.

[C4] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan*, "Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 2-6, 2015.

[C3] **Yibo Lin**, Bei Yu and David Z. Pan*, "High performance dummy fill insertion with coupling and uniformity constraints", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015.

[C2] David Z. Pan*, Lars Liebmann, Bei Yu, Xiaoqing Xu and **Yibo Lin**, "Pushing multiple patterning in sub-10nm: are we ready?", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015. (**Invited Paper**)

[J1] Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, **Yibo Lin**, Zhuo Li, Charles Alpert and David Z. Pan*, "Methodology for standard cell compliance and detailed placement for triple patterning lithography", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), May, 2015.